# Deep Research: Avatrada Ui 9 Topic 001

# Engineering Report: High-Frequency Frontend Architecture

**To**: Lead Engineering Team **From**: Autonomous Technical Researcher **Date**: October 26, 2023 **Subject**: Deep-Dive Analysis of a High-Frequency Trading Frontend using Feature-Sliced Design (FSD) and React 18

## 1. Executive Summary

This report provides a detailed technical specification for building a high-performance, scalable, and maintainable frontend for a financial trading terminal. The architecture is based on the **Feature-Sliced Design (FSD)** methodology for structural integrity and **React 18**'s concurrent features for UI responsiveness under heavy data load.

The analysis deconstructs four key areas: 1. **FSD Folder Structure**: A prescriptive layout for a trading application to ensure modularity and prevent dependency hell. 2. **Architectural Enforcement**: An ESLint configuration to programmatically enforce FSD's strict import boundaries. 3. **High-Throughput State Management**: A custom Redux middleware designed to ingest and batch over 1,000 WebSocket messages per second without compromising a 60fps render loop. 4. **Concurrent UI Rendering**: A practical application of React 18's `useTransition` hook to maintain critical UI responsiveness during non-urgent, heavy state updates.

The findings conclude that this architectural combination is exceptionally well-suited for the demands of a high-frequency trading environment, offering a robust framework for both development velocity and runtime performance.

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# 2. FSD Folder Structure for a Trading Terminal

## 2.1. Technical Deconstruction

Feature-Sliced Design is a prescriptive architectural methodology for frontend applications. Its primary goal is to manage complexity by organizing code into layers and slices, enforcing a unidirectional data and dependency flow. This is critical in a trading terminal where features like "Order Entry," "Charting," and "Position Management" must be developed and maintained in isolation to prevent system-wide failures.

The layers are arranged from most abstract to most specific: - `app`: Application-wide setup (store, routing, global styles). - `processes`: Multi-step user flows (e.g., authentication, multi-leg order execution). - `pages`: Unique application screens, composed of widgets. - `widgets`: Composite UI blocks (e.g., the entire order book). - `features`: User-driven actions (e.g., placing an order). - `entities`: Core business data models (e.g., `Order`, `Instrument`). - `shared`: Reusable, business-agnostic code (UI kits, formatters, API clients).

## 2.2. Implementation Strategy

The following folder structure is tailored for a typical trading terminal. Each slice (e.g., `features/place-order`) exposes a controlled public API via its `index.ts` file.

```
src/
├── app/
│   ├── providers/      # React Context, Redux Provider, Router
│   ├── styles/         # Global styles, theme variables
│   └── index.tsx       # Application entry point
│
├── processes/
│   ├── auth/           # Login, 2FA, logout flow
│   └── trade-execution/ # Multi-stage trade confirmation process
│
├── pages/
│   ├── trading-terminal/ # The main trading dashboard
```

```
│   │   └── index.tsx
│   ├── portfolio/
│   │   └── index.tsx
│   └── settings/
│       └── index.tsx
│
├── widgets/
│   ├── order-book/
│   │   ├── ui/OrderBook.tsx
│   │   └── index.ts      # export { OrderBook } from './ui/OrderBook';
│   ├── price-chart/
│   ├── positions-panel/
│   └── news-feed/
│
├── features/
│   ├── place-order/
│   │   ├── ui/PlaceOrderForm.tsx
│   │   ├── model/orderSlice.ts
│   │   └── index.ts      # export { PlaceOrderForm } from './ui/
PlaceOrderForm';
│   ├── cancel-order/
│   └── switch-instrument/
│
├── entities/
│   ├── order/
│   │   ├── model/types.ts # Order type definitions
│   │   ├── ui/OrderRow.tsx
│   │   └── index.ts      # export * from './model/types'; export
{ OrderRow } ...
│   ├── position/
│   ├── instrument/
│   └── user/
│
└── shared/
    ├── ui/               # Generic components: Button, Input, Spinner
    ├── lib/              # Business-agnostic helpers: formatters, hooks
    ├── api/              # WebSocket client, HTTP client setup
    └── config/           # Environment variables, constants
```

### 2.3. Critical Analysis

- **Failure Mode (Boundary Violation)**: A developer in the `features/place-order` slice might be tempted to directly import the `widgets/positions-panel` to trigger a refresh. This is an illegal import ( `feature` -> `widget` ) that creates tight coupling. The correct approach is for the feature to dispatch a Redux action, and the widget, subscribed to the relevant state, updates independently.
- **Edge Case (Cross-Slice Logic)**: A feature like "Close All Positions" needs to interact with multiple `position` entities. This logic should reside within the feature itself, which orchestrates actions on the `entities` via the store, respecting the FSD flow. The feature acts as the orchestrator, while the entity remains the passive data model.
- **Optimization (Public API)**: The `index.ts` barrel files are critical. They must be curated to expose only what is necessary. Exposing internal state or helper functions breaks encapsulation and makes the slice difficult to refactor. A strict "Public API" discipline is paramount.

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## 3. ESLint Configuration for FSD Boundary Enforcement

### 3.1. Technical Deconstruction

To prevent architectural decay, FSD's layer boundaries must be enforced automatically. Manual code reviews are insufficient and error-prone. The ideal tool for this is a static analyzer like ESLint, configured with rules that restrict import paths between different layers. This turns architectural rules into a CI/CD-enforced reality.

### 3.2. Implementation Strategy

We will use the `eslint-plugin-import` package, specifically its `no-restricted-paths` rule. This allows us to define zones for each FSD layer and specify which other zones they are forbidden from importing. This configuration assumes path

aliases (e.g., `@/features`) are set up in the project's `tsconfig.json` or `jsconfig.json`.

```
// .eslintrc.js

module.exports = {
  // ... other ESLint settings (extends, parser, etc.)
  plugins: ['import'],
  rules: {
    'import/no-restricted-paths': [
      'error',
      {
        zones: [
          // Rule: Shared can't import from anywhere else
          {
            target: './src/shared',
            from: ['./src/entities', './src/features', './src/widgets', './src/
pages', './src/processes', './src/app'],
            message: 'Violation: The `shared` layer cannot import from higher-
level FSD layers.',
          },
          // Rule: Entities can only import from Shared
          {
            target: './src/entities',
            from: ['./src/features', './src/widgets', './src/pages', './src/
processes', './src/app'],
            message: 'Violation: The `entities` layer cannot import from higher-
level FSD layers.',
          },
          // Rule: Features can only import from Entities and Shared
          {
            target: './src/features',
            from: ['./src/widgets', './src/pages', './src/processes', './src/
app'],
            message: 'Violation: A `feature` cannot import from `widgets`,
`pages`, `processes`, or `app`.',
          },
          // Rule: Widgets can only import from Features, Entities, and Shared
          {
            target: './src/widgets',
```

```
            from: ['./src/pages', './src/processes', './src/app'],
            message: 'Violation: A `widget` cannot import from `pages`,
`processes`, or `app`.',
          },
          // Rule: Pages can only import from Widgets, Features, Entities, and
Shared
          {
            target: './src/pages',
            from: ['./src/processes', './src/app'],
            message: 'Violation: A `page` cannot import from `processes` or
`app`.',
          },
          // Rule: Processes can import from any layer except App
          {
            target: './src/processes',
            from: ['./src/app'],
            message: 'Violation: A `process` cannot import from `app`.',
          },
        ],
      },
    ],
  },
  settings: {
    'import/resolver': {
      typescript: { // or 'node' if using JS with jsconfig
        alwaysTryTypes: true,
      },
    },
  },
};
```

## 3.3. Critical Analysis

- **Failure Mode (Configuration Drift)**: If a new layer is added or path aliases change, this ESLint configuration must be updated. Failure to do so will render the rules ineffective or cause false positives. This file should be considered a core architectural artifact.

- **Edge Case (Type Imports)**: Sometimes, a lower layer may need to import a TypeScript `type` from a higher layer without importing any runtime code. The `import type { ... } from '...'` syntax can sometimes bypass simple path restrictions depending on the parser. More advanced static analysis or custom rules might be needed if this becomes a problem, but `no-restricted-paths` is generally sufficient.
- **Optimization (IDE Integration)**: For maximum effectiveness, the ESLint plugin must be integrated into the developers' IDEs (e.g., VS Code ESLint extension). This provides immediate feedback, preventing boundary violations before the code is even committed.

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# 4. High-Frequency Redux Middleware for WebSocket Data

## 4.1. Technical Deconstruction

A trading terminal's primary data source is often a WebSocket connection streaming market data (ticks). A naive implementation would dispatch a Redux action for every incoming message. At 1,000+ messages/sec, this would trigger 1,000+ state updates and re-renders, overwhelming the React reconciler and freezing the browser's main thread.

The solution is to create a custom Redux middleware that acts as a buffer. It intercepts incoming tick actions, collects them, and dispatches a single, batched update action synchronized with the browser's paint cycle using `requestAnimationFrame`. This ensures the UI updates at a maximum of 60fps (or the monitor's refresh rate), regardless of the incoming message frequency.

## 4.2. Implementation Strategy

This middleware buffers ticks in a `Map` to ensure only the latest price for each instrument is stored within a single frame. It uses a flag (`isUpdateScheduled`) to prevent scheduling more than one `requestAnimationFrame` callback per frame.

```
// src/shared/api/websocket-middleware.ts

import { Middleware, Dispatch, AnyAction } from '@reduxjs/toolkit';
import { marketSlice } from '../../entities/instrument/model/marketSlice';

// A Map is used to store the latest tick for each symbol, overwriting older
ones.
const tickBuffer = new Map<string, number>();
let isUpdateScheduled = false;

export const websocketTickBufferingMiddleware: Middleware =
  (store) => (next: Dispatch) => (action: AnyAction) => {
    // Intercept only the specific raw tick action from the WebSocket client
    if (action.type !== 'websocket/rawTickReceived') {
      return next(action);
    }

    const { symbol, price } = action.payload;
    tickBuffer.set(symbol, price);

    // If an update is not already scheduled for the next frame, schedule one.
    if (!isUpdateScheduled) {
      isUpdateScheduled = true;

      requestAnimationFrame(() => {
        // Dispatch a single batched action with all collected ticks

store.dispatch(marketSlice.actions.updateTicks(Object.fromEntries(tickBuffer)));

        // Clear the buffer and reset the flag for the next frame
        tickBuffer.clear();
        isUpdateScheduled = false;
      });
    }
  };

// Example usage in Redux store configuration:
// configureStore({
//   reducer: { ... },
//   middleware: (getDefaultMiddleware) =>
```

```
//     getDefaultMiddleware().concat(websocketTickBufferingMiddleware),
// });

// In the marketSlice reducer (using RTK's Immer for efficient mutation):
// updateTicks: (state, action: PayloadAction<Record<string, number>>) => {
//   for (const symbol in action.payload) {
//     if (state.prices[symbol]) {
//       state.prices[symbol] = action.payload[symbol];
//     }
//   }
// }
```

## 4.3. Critical Analysis

- **Failure Mode (Main Thread Blocking)**: If another part of the application blocks the main thread for an extended period (e.g., > 100ms), the `requestAnimationFrame` callback will be delayed. During this time, the `tickBuffer` could accumulate a large number of messages, leading to a memory spike upon the thread's release. A potential mitigation is to cap the buffer size, though this would mean dropping data.
- **Edge Case (Tab Throttling)**: Browsers heavily throttle `requestAnimationFrame` for background tabs. This is generally desired behavior, as it saves resources. However, if the application needs to maintain a precise state log even when backgrounded, this middleware is insufficient. A Web Worker would be required to process the WebSocket stream off the main thread.
- **Optimization (Data Structure)**: Using a `Map` is highly efficient for this use case. It provides O(1) insertion and update complexity, and it naturally de-duplicates ticks for the same instrument within a single animation frame, ensuring only the most recent data is processed.

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# 5. `useTransition` for UI Responsiveness

## 5.1. Technical Deconstruction

React 18 introduces concurrency, allowing React to work on multiple state updates simultaneously. The `useTransition` hook is a key tool for leveraging this. It allows us to mark specific state updates as "non-urgent." React can then interrupt the rendering of these non-urgent updates to handle more critical, "urgent" updates, such as user input.

In a trading terminal, an "Emergency Stop" button click is an urgent update that must be instantaneous. In contrast, updating a large, complex news feed is a non-urgent update that can be deferred or interrupted if the user performs a critical action.

## 5.2. Implementation Strategy

In this example, a `TradingDashboard` component receives news updates. We wrap the state update for the news feed in `startTransition`. This tells React that rendering the new news feed can be de-prioritized. The `onClick` handler for the "Emergency Stop" button updates state directly, making it an urgent update that will interrupt the news feed render if necessary.

```
// src/widgets/trading-dashboard/ui/TradingDashboard.tsx

import React, { useState, useTransition, useEffect } from 'react';
import { NewsFeed } from '../../news-feed';
import { EmergencyStopButton } from '../../../features/emergency-stop';

// Mock API to simulate incoming news data
const mockNewsApi = {
  subscribe: (callback: (data: any[]) => void) => {
    setInterval(() => callback(generateHeavyNewsData()), 2000);
  }
};

function TradingDashboard() {
  const [newsItems, setNewsItems] = useState([]);
```

```
  const [isPending, startTransition] = useTransition();

  useEffect(() => {
    // Subscribe to news updates
    mockNewsApi.subscribe(newNewsData => {
      // This is a non-urgent update. It can be interrupted.
      startTransition(() => {
        setNewsItems(newNewsData);
      });
    });
  }, []);

  const handleEmergencyStop = () => {
    // This is an URGENT update. It will interrupt the news feed render.
    console.log('EMERGENCY STOP INITIATED - IMMEDIATE ACTION');
    // ... logic to dispatch stop action
  };

  return (
    <div>
      <h1>Trading Dashboard</h1>
      <EmergencyStopButton onClick={handleEmergencyStop} />

      <hr />

      <h2>Live News Feed</h2>
      {isPending && <div className="spinner">Updating news...</div>}
      <NewsFeed items={newsItems} />
    </div>
  );
}
```

## 5.3. Critical Analysis

- **Failure Mode (Incorrect Usage)**: If `startTransition` is used to wrap an urgent update (like a controlled input's `onChange`), it will introduce noticeable input lag, degrading the user experience. `useTransition` must be applied selectively only to state updates that can tolerate a delay.

- **Edge Case (Stale Data Perception)**: During a transition, the UI can be in a state where some parts have updated (urgent changes) while others have not (the pending transition). The `isPending` state is crucial for providing user feedback (e.g., a loading spinner or dimmed view) to clearly indicate that a background update is in progress, preventing user confusion.
- **Optimization (`useDeferredValue`)**: For cases where you don't control the state update logic (e.g., a value comes from a parent component's props), `useDeferredValue` provides a similar de-prioritization mechanism. It can be used to defer rendering a computationally expensive component based on a prop value, preventing it from blocking more critical UI elements.